AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2014

Mathematics Paper II

Time: 2 hours 20 minutes Marks: 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are ELEVEN questions. Answer ALL questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

(ATTEMPT EITHER PART a OR PART b OF Q.1.)

Q.1. (Total 5 Marks)

a. Simplify $\left(\frac{1}{x^3 - x} + \frac{1}{x^2 - 1}\right) \div \frac{1}{(x-1)^2}$ (5 Marks)

b. Find:

i. The square root of $x^4 - 8x^2 + 16$ (3 Marks)

ii. The highest common factor (HCF) of $x^3 - 64$ and $(x-4)^2$ (2 Marks)

Q.2. (Total 3 Marks)

Match correctly each item in column A with an item in column B by an arrow $(\rightarrow)$.

Column A	Column B
$\frac{x}{x^2-1}$	$\frac{A}{(x+1)}+\frac{B}{(x+1)^2}+\frac{C}{x-1}$
$\frac{x}{(x+1)^2(x-1)}$	$\frac{A}{(x+1)^2}+\frac{B}{(x-1)}$
$\frac{x}{(x+2)(x+1)}$	$\frac{A}{x-1}+\frac{B}{x+1}$
	$\frac{A}{x+2}+\frac{B}{x+1}$

(ATTEMPT EITHER PART a OR PART b OF Q.3.)

Q.3. (Total 5 Marks)

a. Solve $\frac{\sqrt{x+c}}{c} - c = c$ and verify its solution.

b.

i. Solve $3x + \frac{2}{3} = \frac{8}{3}$ (2 Marks)

ii. Find the solution set of $3 - 2x < 5, x \in R$. Also represent the solution set on the number line. (3 Marks)

Q.4. (Total 4 Marks)

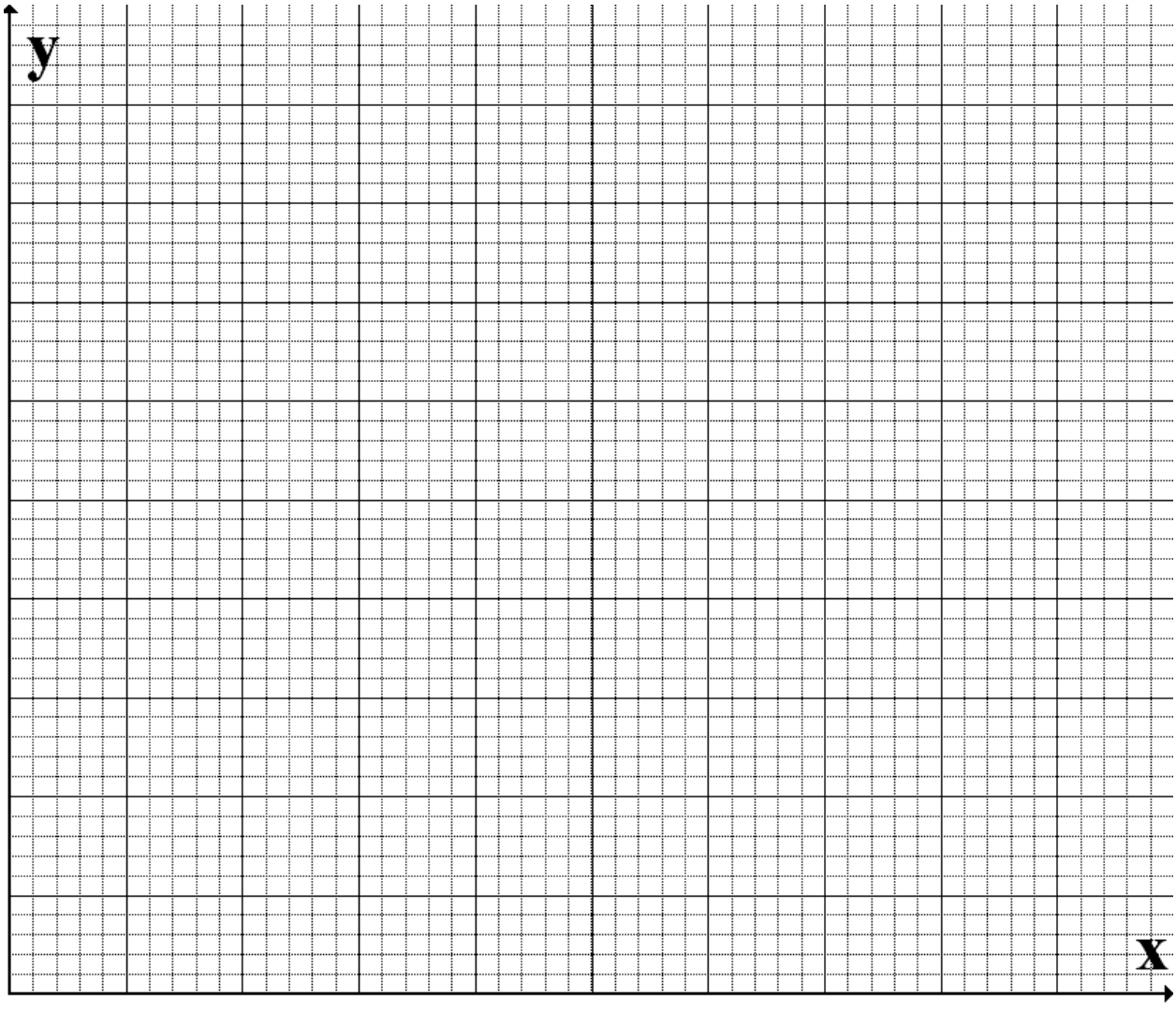
For the given simultaneous linear equations complete the following tables and find the point of intersection by the graphical method.

x	0	1	
y	6		0

$3x + y = 6$

x	10		7
y	0	2	

$x + 3y = 10$



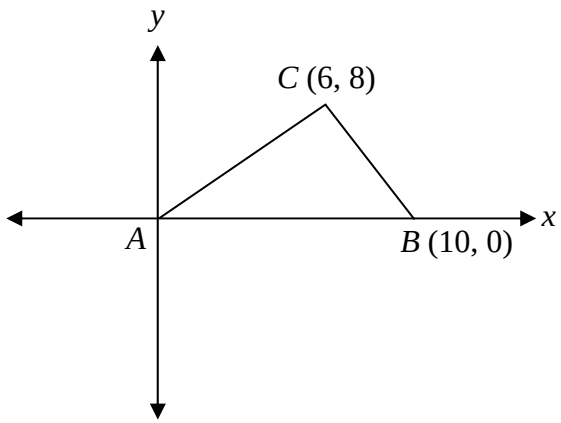
Q.5. (Total 4 Marks)

Find the values of x for $6x^2 - 31x + 35 = 0$

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

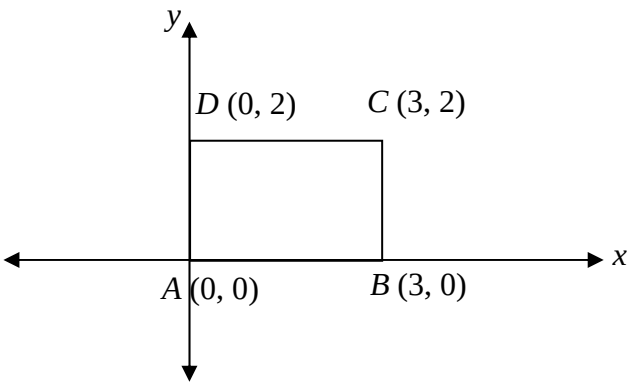
Q.6. (Total 3 Marks)

a. Prove that in the given diagram, ΔABC is an isosceles triangle.



NOT TO SCALE

b. Consider the given diagram.



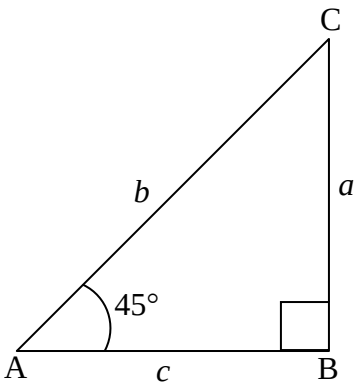
Find the mid-points of the diagonals and show that the diagonals have the same mid-point.

(ATTEMPT EITHER PART a OR PART b OF Q.7.)

Q.7. (Total 4 Marks)

a. Prove that $\left(1+\tan^2\theta\right)\left(1-\cos^2\theta\right)=\tan^2\theta$

b. Using the given figure, calculate the values of $\sin 45°$ and $\tan 45°$

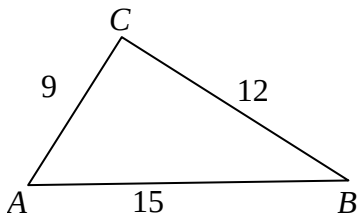


(ATTEMPT EITHER PART a OR PART b OF Q.8.)

Q.8. (Total 4 Marks)

a. Show that the sides $(a - b)$, $2\sqrt{ab}$ and $(a + b)$ represent a right angled triangle, where a and b are positive integers. [**Note**: $a > b$]

b. Prove that ΔABC is a right angled triangle. Identify the right angle of the triangle.

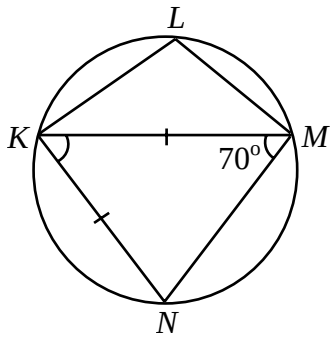


Q.9. (Total 4 Marks)

Prove by drawing that when two circles touch each other externally or internally, the distance between their centres is equal to the sum or difference of their radii respectively.

Q.10. (Total 4 Marks)

For the given figure, find



i. the sum of angles $K\,L\,M$ and $K\,N\,M$. Also give a reason to support your answer.

ii. the angle $M\,K\,N$.

Q.11. (Total 5 Marks)

Construct a triangle ABC with $m\overline{BC} = 6\,cm$, $m\angle B = 45°$ and $m\angle C = 60°$. Also draw the circumscribed circle.

END OF PAPER